

B.Tech. Degree III Semester Examination November 2012

SE 304 MANUFACTURING PROCESS (2006 Scheme)

Time : 3 Hours

Maximum Marks : 100

PART A (Answer *ALL* questions)

(8 × 5 = 40)

- I. (a) How engineering materials are classified?
(b) What is meant by property of a material? What are the important mechanical properties?
(c) Cite the differences between TIG and MIG.
(d) What is called as weldability? Explain the merits and demerits of welding.
(e) What is pattern? What are the important factors upon which the selection of pattern material depend?
(f) Describe about the investment casting procedure.
(g) Differentiate orthogonal cutting and oblique cutting.
(h) What are the different types of rolling mills?

PART B

(4 × 15 = 60)

- II. (a) With a neat diagram, explain Iron-Carbon equilibrium. What are its equilibrium phase transformations? Explain about its critical points. (10)
(b) What are the thermal properties of a material? (5)
- OR**
- III. (a) Explain the different non-ferrous metals and alloys, its properties, process and uses. (9)
(b) Describe the following: (6)
(i) Annealing (ii) Tempering.
- IV. (a) Explain the different arc welding processes. (9)
(b) What are the differences between spot welding and seam welding? (6)
- OR**
- V. (a) With neat diagrams, explain the weld defects. (9)
(b) Write short notes on: (6)
(i) Electron beam welding
(ii) Laser beam welding
- VI. (a) What are the factors responsible for a casting defect? Explain the different casting defects and its inspection methods. (10)
(b) Explain the shell moulding procedure. What are its advantages? (5)
- OR**
- VII. (a) How moulding sands are classified? What are the properties of it? (9)
(b) Explain the different types of cores. (6)
- VIII. (a) What is an extrusion process? What are the different methods of extrusion? (9)
(b) Explain the different cold working processes. (6)
- OR**
- IX. (a) Describe the advanced machining methods in detail. (10)
(b) What is meant by hot working? Explain the advantages and disadvantages of hot working. (5)